

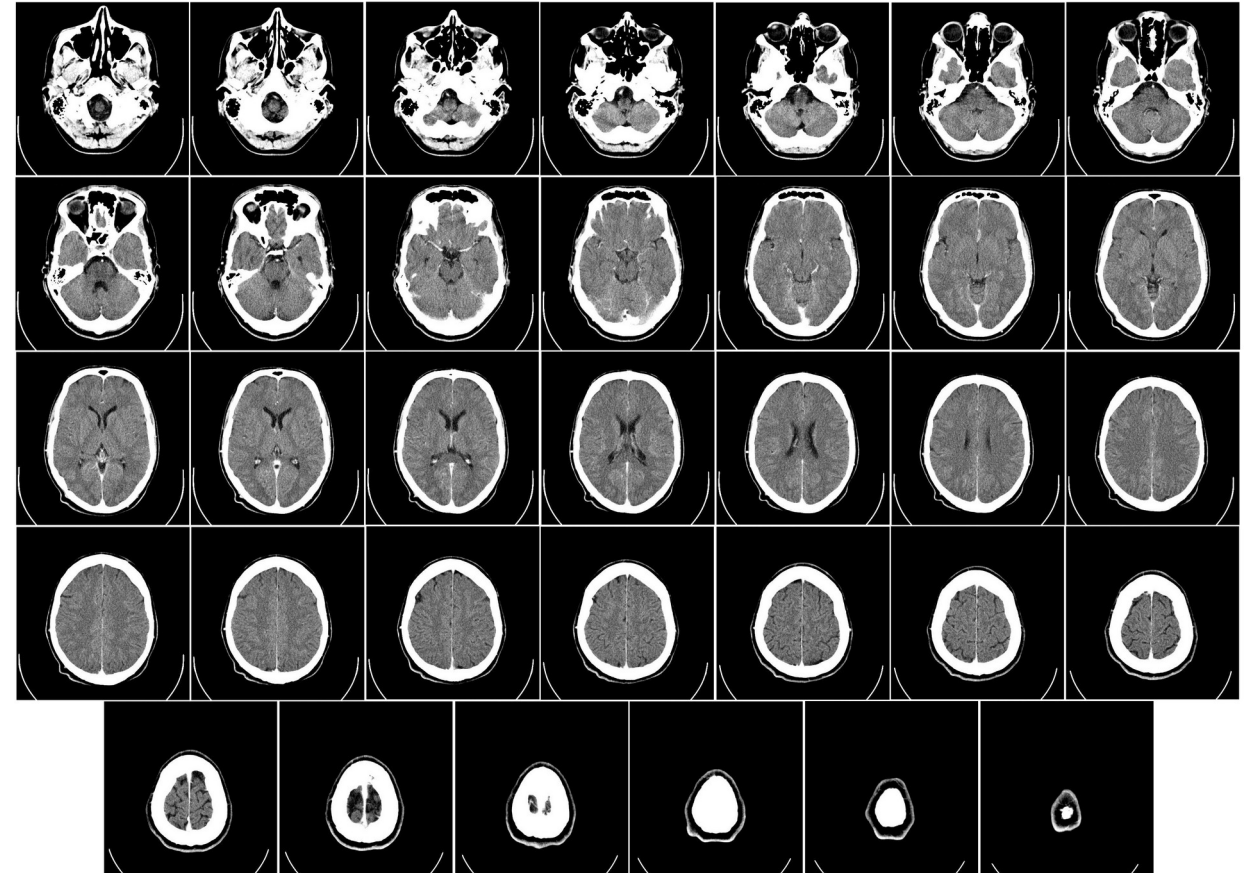
Course: TPG4190, NTNU

Bhargav Boddupalli
Research Scientist
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What is Tomography?

- Tomography is imaging by sections or sectioning that uses any kind of penetrating wave. The method is used in radiology, archaeology, biology, atmospheric science, geophysics, oceanography, plasma physics, materials science, cosmochemistry, astrophysics, quantum information, and other areas of science.
- For example: A computed tomography scan (CT scan) is a medical imaging technique used to obtain detailed internal images of the body.



Computed tomography of human brain, from base of the skull to top. Source: Wikipedia



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Seismic Tomography

- **Seismic tomography** is the art of translating recordings of seismic waves into quantitative images of the Earth's interior.
- **Why do we need seismic tomography?**
- Seismic tomography reveals the dynamics and composition of the Earth and other planetary bodies, illuminates the structure of active volcanoes and fault zones, constrains the internal flow patterns of glaciers and ice sheets, provides information on deep reservoirs and storage sites, and enables the accurate simulation of earthquake-induced ground motion, to list just a few of its most prominent uses.



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Seismic refraction tomography

- Uses first-arrival seismic waves to estimate subsurface P-wave velocity structure.
- Measures travel times of refracted waves recorded at multiple receivers.
- Produces a 2D or 3D velocity model of the subsurface.
- Zelt and Barton, 1998 – First arrival traveltimes tomography (FATT)

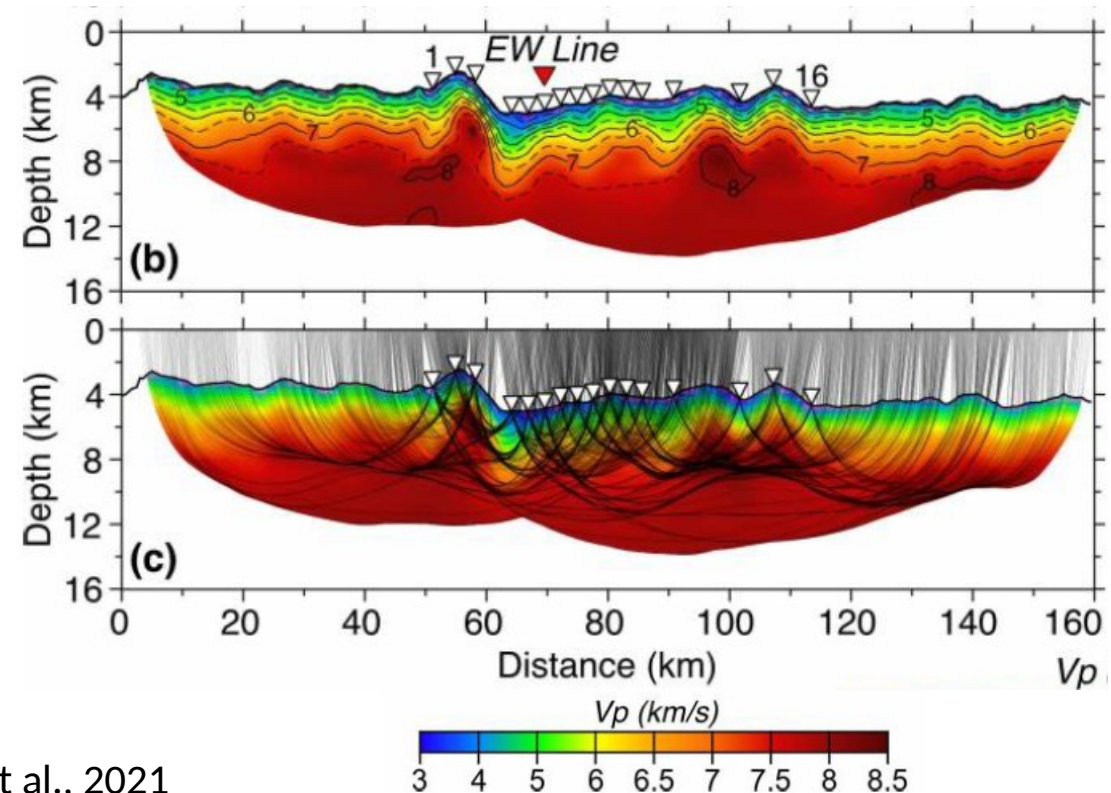
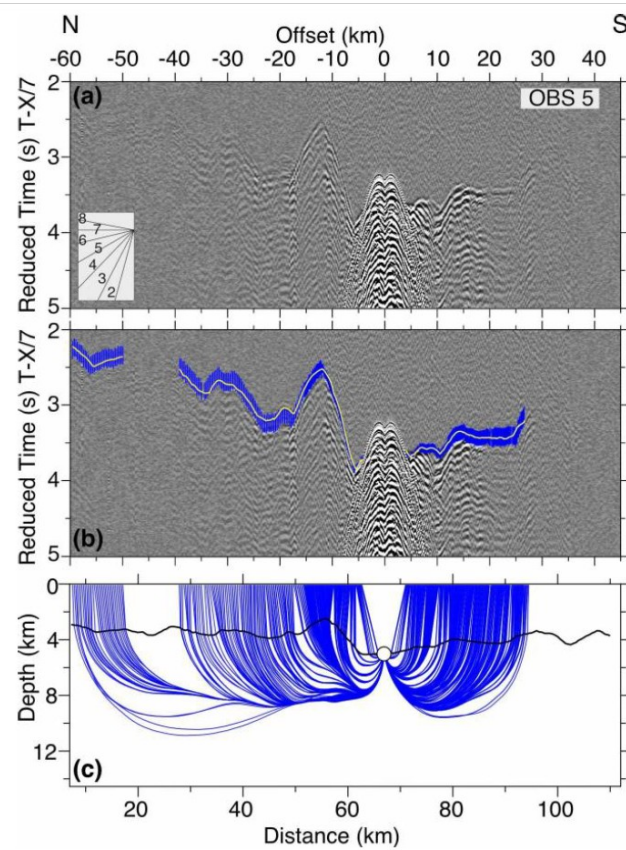
Seismic data



Velocity model

Seismic refraction tomography

- Record observed travel times from seismic data.
- Predict theoretical travel times using a starting velocity model (forward modeling).
- Adjust model until calculated and observed times agree (inversion).



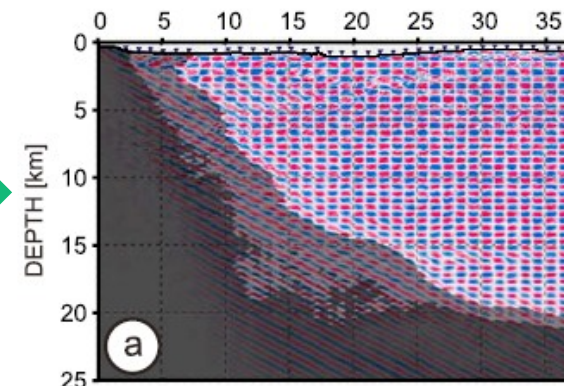
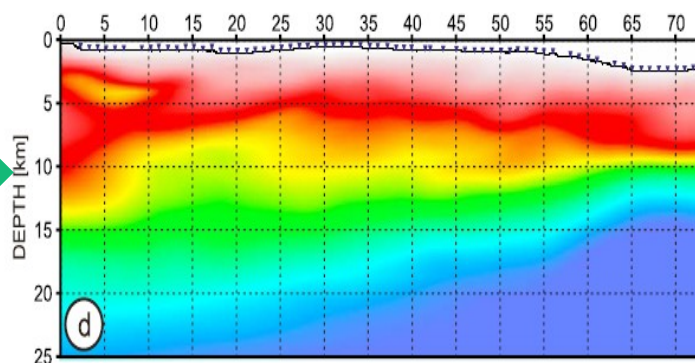
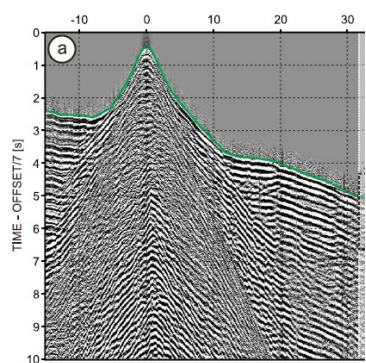
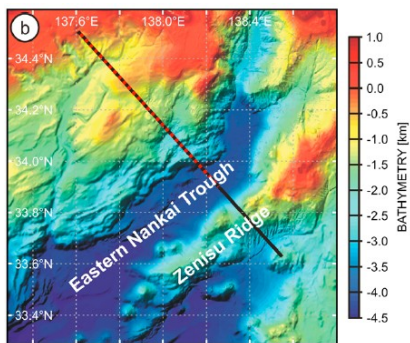


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Components of Seismic Refraction Tomography

- **Data acquisition** – shots and receivers (OBS or land geophones).
- **Travel-time picking** – identify first arrivals on seismograms.
- **Forward modeling** – compute travel times through a model.
- **Inversion** – update model to minimize misfit.
- **Quality control & resolution testing.**



Forward modelling

- Forward modeling in tomography is based on the Eikonal equation, which comes from the high-frequency (ray) approximation of the wave equation:

$$|\nabla T(\mathbf{x})| = \frac{1}{v(\mathbf{x})}$$

$$|\nabla T|^2 = s(\mathbf{x})^2$$

$T(\mathbf{x})$

Traveltime from the source to point $\mathbf{x} = (x, y, z)$

$v(\mathbf{x})$

Velocity at point $\mathbf{x}$

$s(\mathbf{x}) = 1/v(\mathbf{x})$

Slowness

∇T

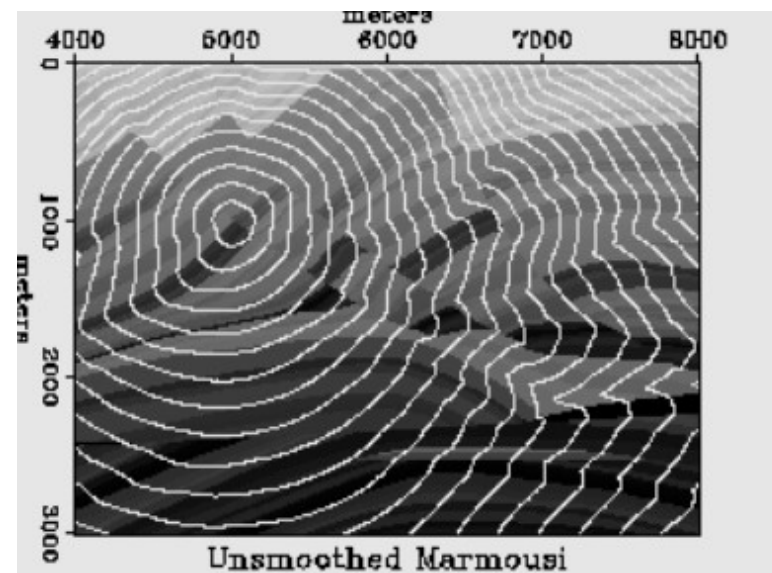
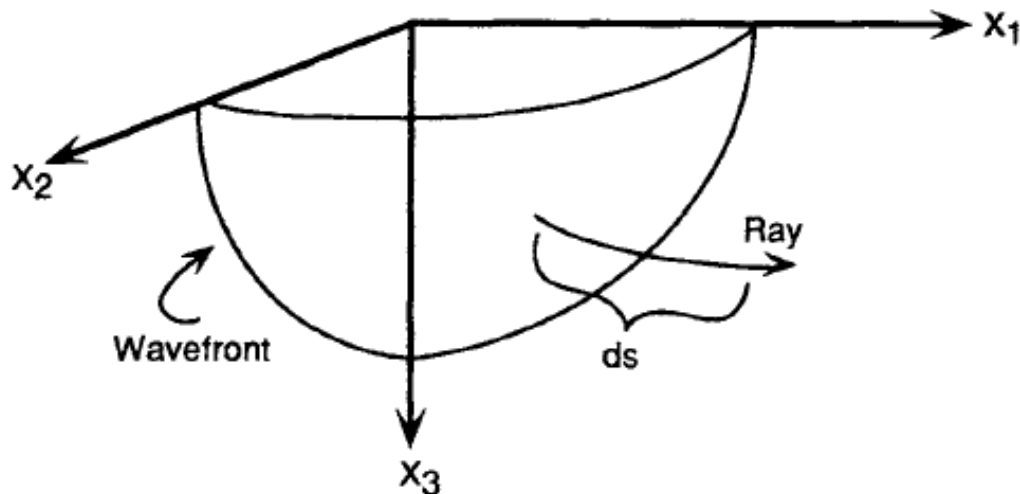
Gradient of traveltime, i.e. direction of wave propagation



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Constant Travel Time Contours



- It describes how a wavefront (surface of equal travel time) moves through space.
- The gradient of travel time gives the ray direction — perpendicular to the wavefront.
- In regions of low velocity, v is small $\rightarrow s$ large $\rightarrow$ wavefronts bend toward that region (Snell's Law behavior).

Inversion

- Predicts first-arrival travel times through a velocity model.
- Solves the Eikonal equation for ray traces
- Used repeatedly during inversion.

$$\text{misfit} = \sum (t_{\text{obs}} - t_{\text{calc}})^2$$

- The model is updated iteratively until misfit is minimized.
- Two inversion styles:
 - Back projection
 - Regularized inversion



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Backpropagation

$$\Delta t = t_{\text{obs}} - t_{\text{calc}}$$

- **Basic Idea:** Each ray that connects a source and receiver provides information about the average velocity along its path.
- If a ray arrives later than predicted → the model is too slow along that path.
- If a ray arrives earlier → the model is too fast.
- The simplest way to fix the model: “Spread” the time residual (Δt) back along the ray path, adjusting slowness in each cell the ray crosses.
- That’s why it’s called back projection — you project the travel-time residuals back into the model.
- This back propagation happens in iterations updating the model at each iteration



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Regularized Inversion

- **Basic Idea:** Here the inversion is treated as a mathematical optimization problem:
- Find the smoothest velocity model that fits the observed data within its uncertainty.
- It minimizes an objective (cost) function that combines two terms:

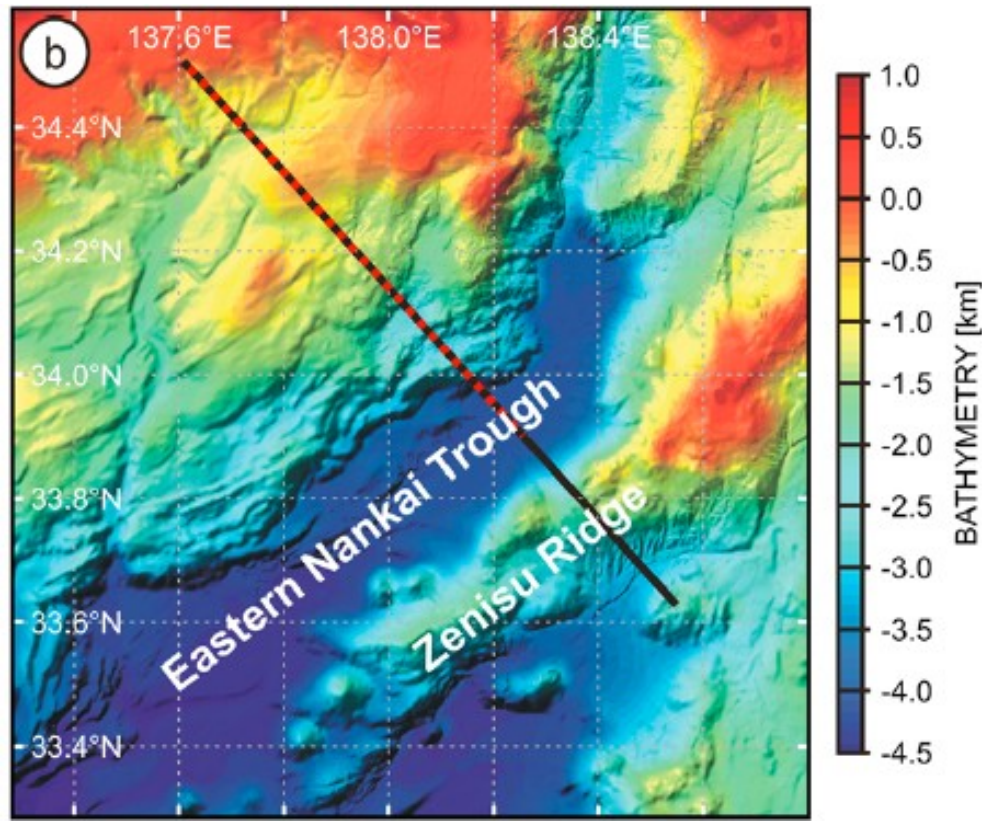
$$\Phi(m) = \text{Data Misfit} + \lambda \times \text{Model Roughness}$$

where:

- m = vector of model parameters (e.g., slowness values),
 - λ = trade-off parameter that balances data fit and smoothness.
- Linearized around the current model: $t=Lm$, where L is the matrix of partial derivatives (path lengths of rays through each cell).
 - Solve a large system of equations using conjugate gradient methods (e.g., LSQR).
 - Update model iteratively: Compute new travel times (forward modeling). Recalculate residuals. Repeat until misfit ≈ 1 (fits data within uncertainty).

Example of First arrival travel time Tomography

Gorszczyk et al. (2017)



- **Survey context**

- Region: Eastern Nankai Trough, Tokai segment (Japan)
- Aim: Deep crustal imaging of subduction zone structures
- Acquisition: 2001 KY0106 cruise (R/V Kaiyo) under the SEIZE France–Japan (SFJ) project

- **OBS configuration**

- 100 Ocean Bottom Seismometers (OBS)
- Spacing: **1 km**
- Profile length: **100 km**
- Vertical 3-component geophones + hydrophones

- **Source configuration**

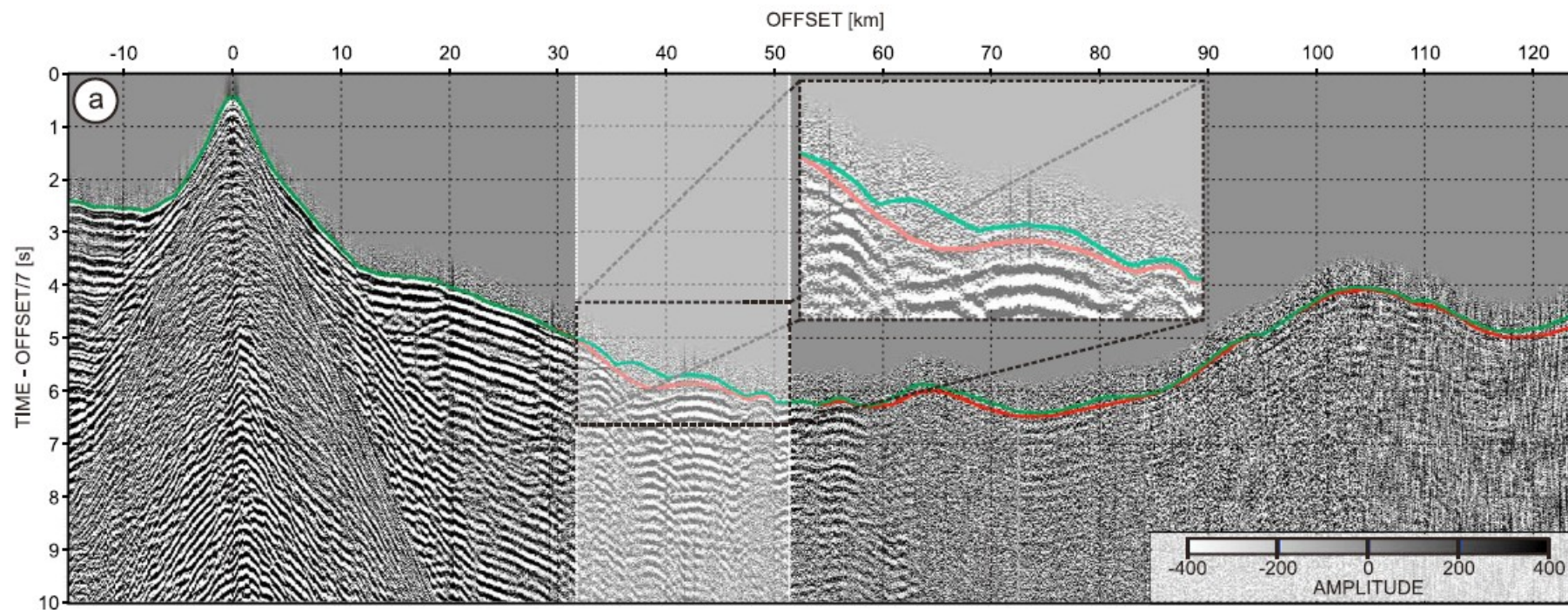
- **1404 air-gun shots**, spacing **100 m**
- Total source line length: **140 km**
- Air-gun array: **196 L total volume**



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OBS Data



- The red and green lines represent the initial and final first-arrival travel time picks, respectively.

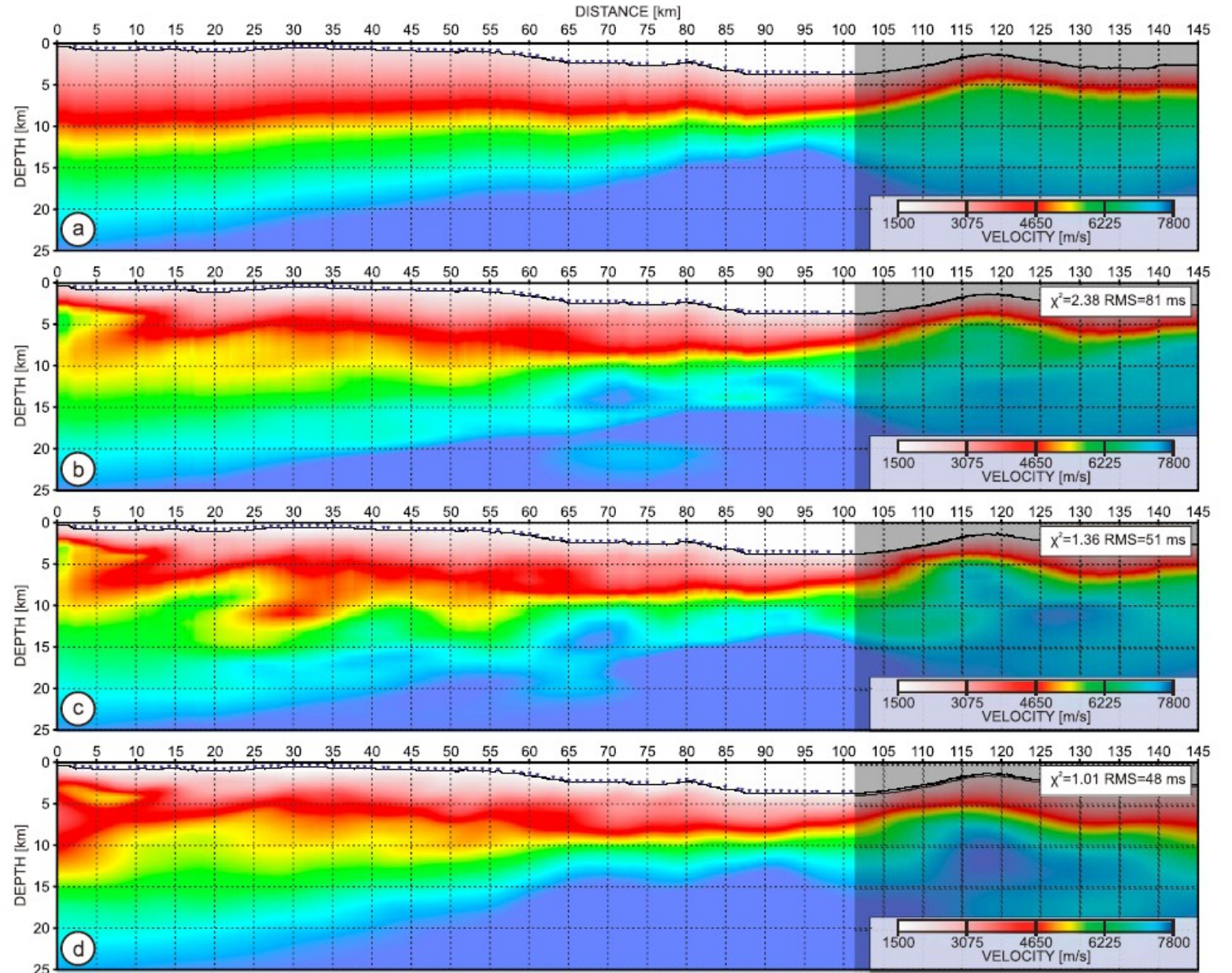


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Result from FATT

- FATT performed in multiple iterative stages





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Questions!

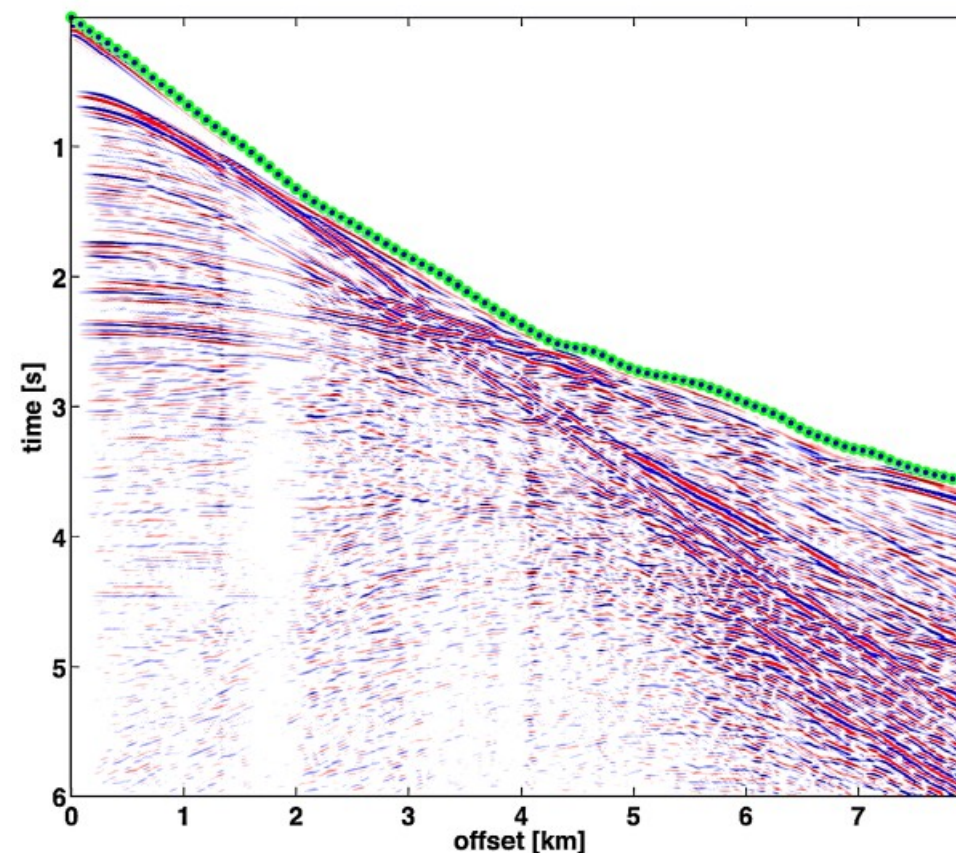


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Full Waveform Inversion

- FWI is a high-resolution imaging technique that is based on modelling the complete waveform for extracting the physical parameters of the medium.
- Traveltime tomography uses only the arrival times
- FWI will invert for complete waveform



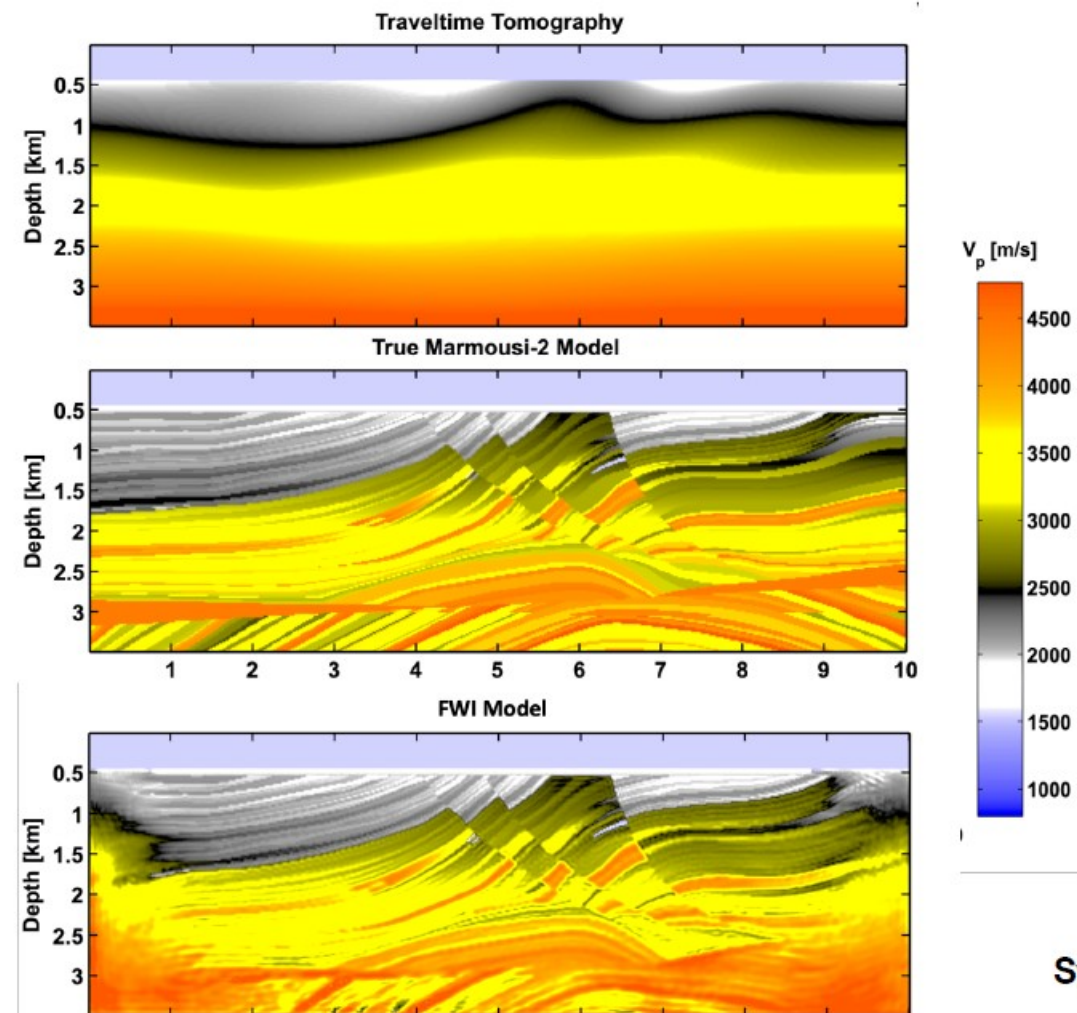


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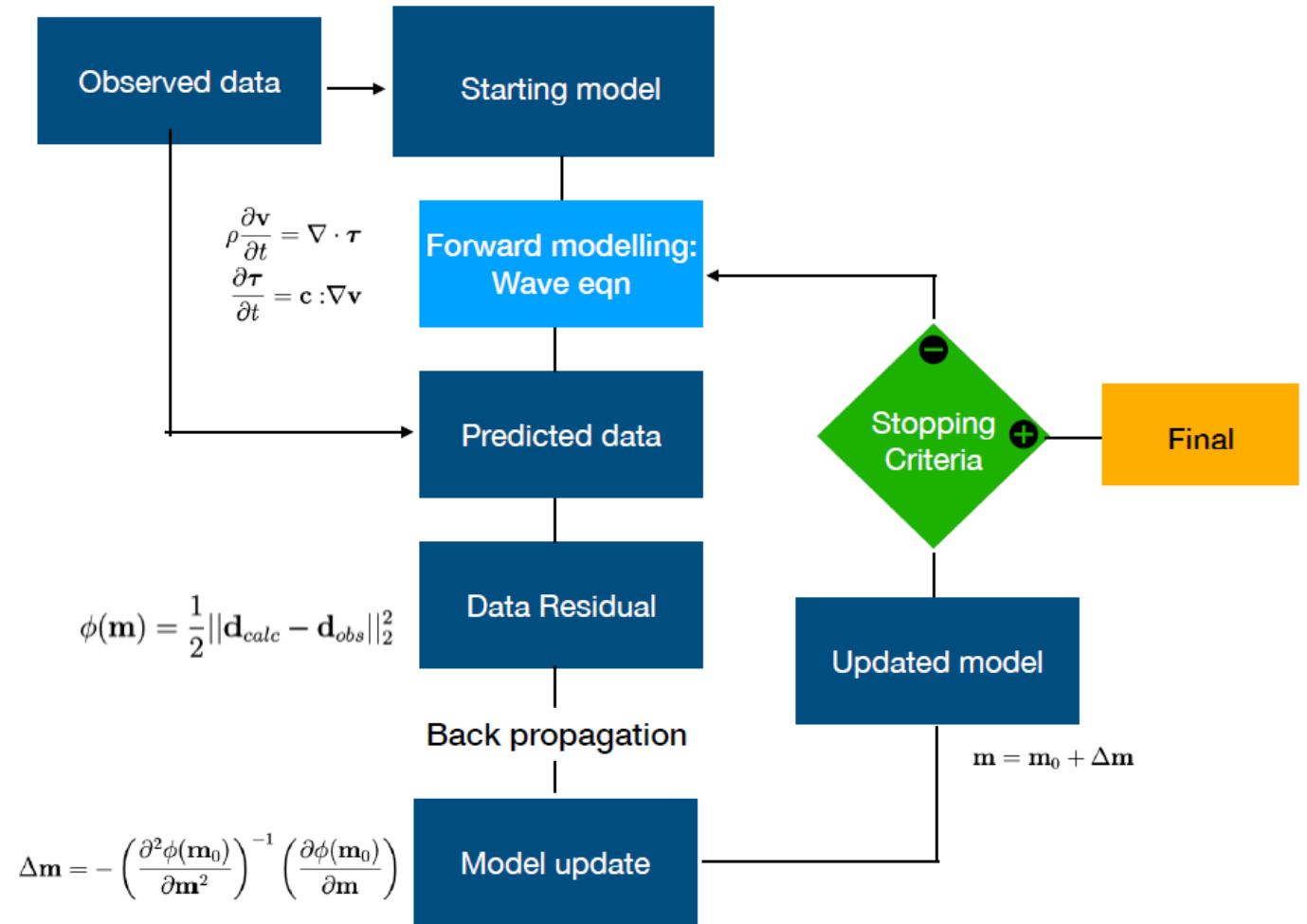
Full Waveform Inversion

- FWI iteratively minimizes the difference between field data and synthetic data
- Requires a starting model – usually developed from traveltime tomography



How does FWI work?

- Main components:
 - A starting model
 - Forward modelling operator
 - Optimization



Forward modelling

- Seismic modelling provides the forward engine for FWI.
- Objective: accurately simulate wave propagation in the Earth.
- Key parameters: **V_p**, **V_s**, **ρ**, and geometry.

Acoustic Wave Equation

$$\frac{\partial^2 p}{\partial t^2} = V_P^2 \nabla \cdot \left(\frac{1}{\rho} \nabla p \right) + s$$

- p : acoustic pressure
- V_P : compressional (P-wave) velocity
- ρ : density
- s : external source (e.g., air gun or vibrator)

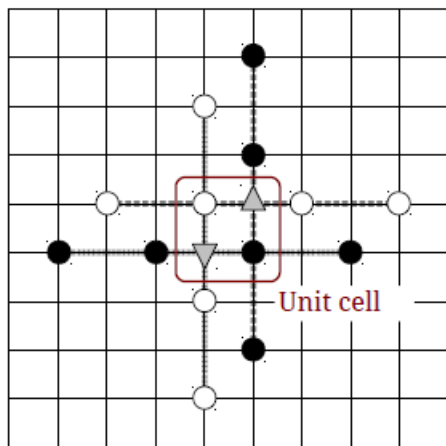
Elastic Wave Equation

$$\rho \frac{\partial^2 u_i}{\partial t^2} = \partial_j \tau_{ij} + f_i, \quad \tau_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij}$$

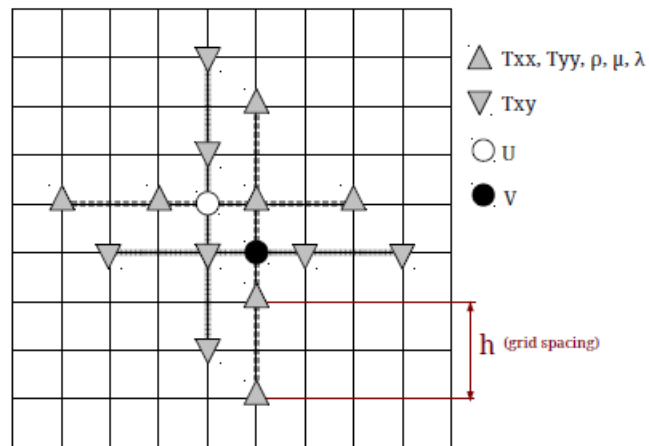
- u_i : particle displacement (components $i = x, y, z$)
- τ_{ij} : stress tensor
- ε_{ij} : strain tensor
- λ, μ : Lamé parameters (control compressional & shear rigidity)
- f_i : body-force source term (e.g., seismic source)
- δ_{ij} : Kronecker delta

Forward modelling -- Numerical Solution Methods

- Three main families:
- Finite Differences (FD) – simple, regular grid.
- Finite Elements (FE) – flexible geometry.
- Spectral Methods – high accuracy, frequency domain.

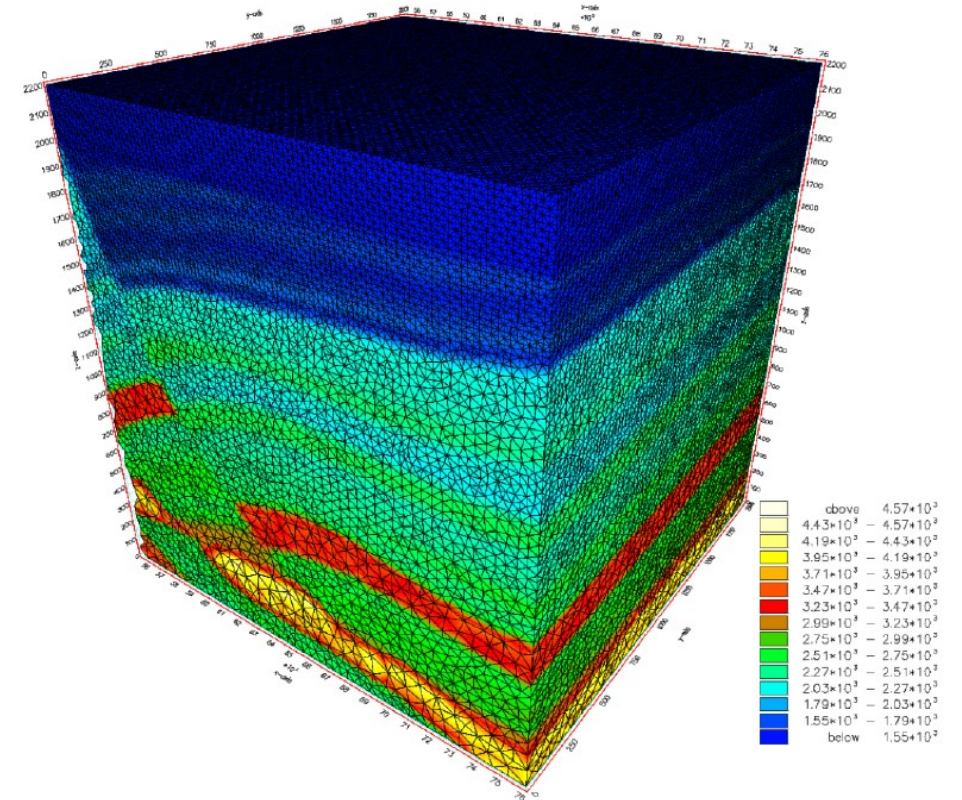


(a) stresses



(b) velocities

Finite difference stencil



Finite element mesh

Forward modelling -- Finite difference Operators

- **Principle:** Replace derivatives in the wave equation with finite-difference (FD) approximations on a discrete grid.
- Central difference for 2nd-order accuracy. Higher-order schemes (4th, 6th, 8th order) improve accuracy and reduce dispersion.

Spatial derivative:

$$\frac{\partial f}{\partial x} \approx \frac{f(x + \Delta x/2) - f(x - \Delta x/2)}{\Delta x}$$

Temporal derivative:

$$\frac{\partial^2 f}{\partial t^2} \approx \frac{f(t + \Delta t) - 2f(t) + f(t - \Delta t)}{\Delta t^2}$$



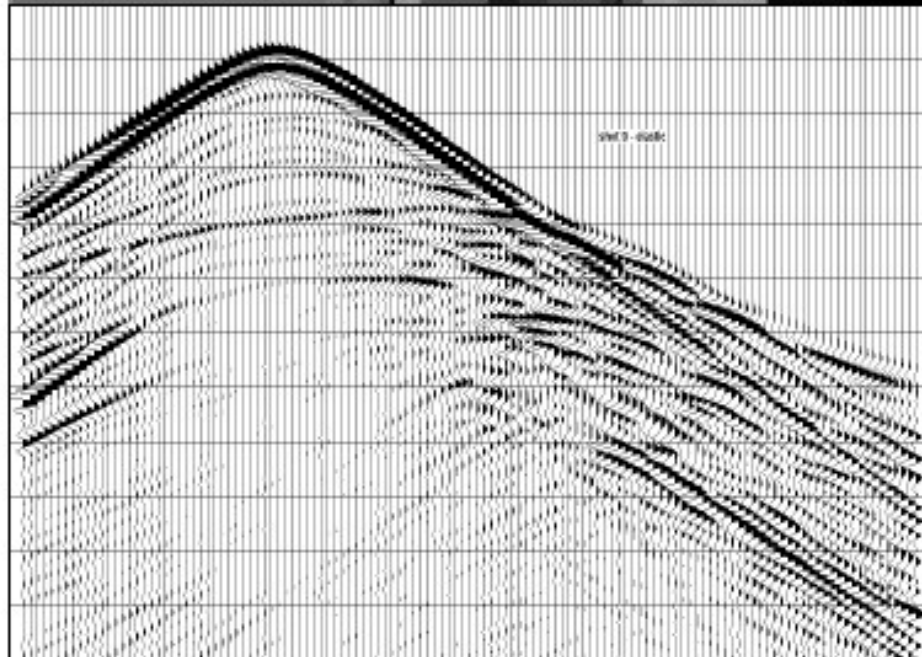
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Forward modelling -- Elastic wave propagation



Snapshot of the wavefield propagating through the model



Seismic data recorded at receivers



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Starting Model

- The starting model should be able to predict the data within half a wavecycle to prevent cycle skipping

Boddupalli et al. (2021)

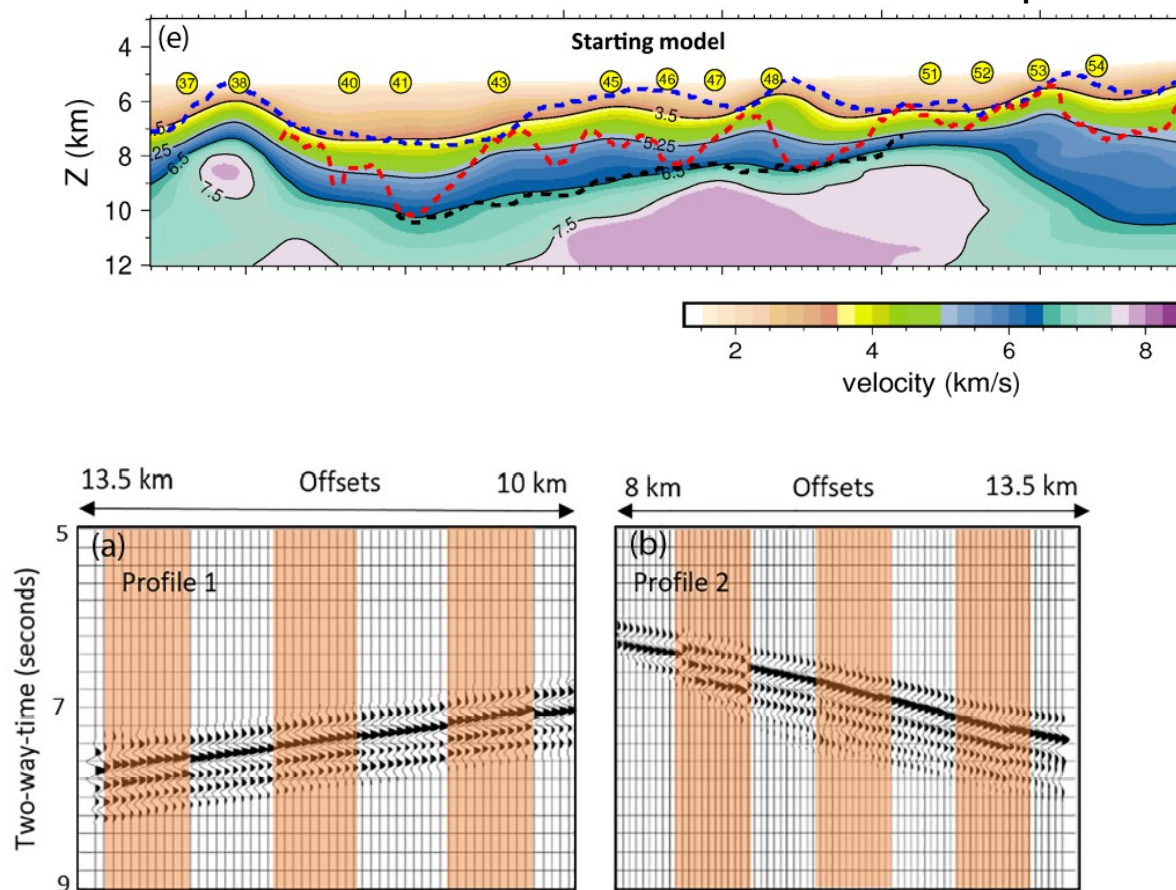


Figure 4. Observed (highlighted) and predicted data using the starting model plotted in sets of 10 traces from OBS 46, alternating to show the match between the arrivals along two profiles. (a) Profile 1 is the closest profile to the OBS 46 location and (b) Profile 2 is at a distance of ~ 10 km from this location. The data shown are bandpass filtered and trace-by-trace normalized.



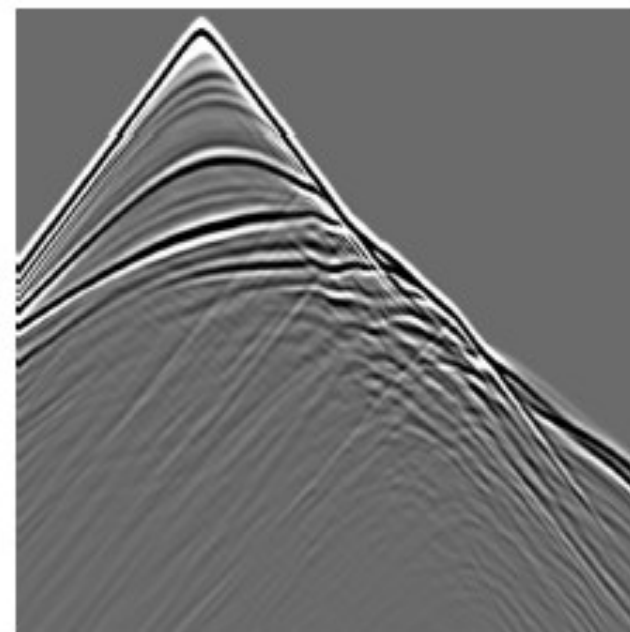
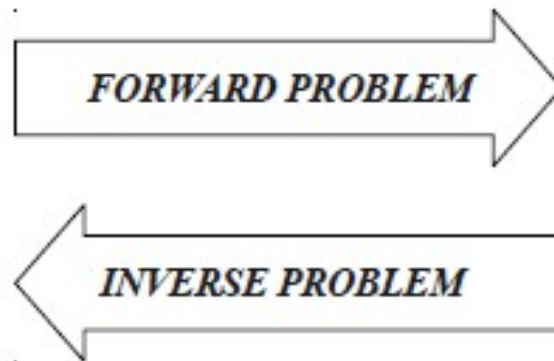
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Inverse Problem



model parameters ($\mathbf{m}$)



data($\mathbf{d}$)



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Inverse Problem

- **Forward problem:**

Predict data d_{mod} from a known model m :

$$d_{mod} = G(m)$$

where G is the forward operator (e.g., wave propagation).

- **Inverse problem:**

Estimate model m from observed data d_{obs} .

- **Objective:** Find model parameters that minimize the mismatch between modelled and observed data.
- Inverse problems are often:
 - **Non-linear** (data not linearly related to parameters)
 - **Ill-posed** (non-unique or unstable)



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Inverse Problem

- Data misfit (least squares objective) function:

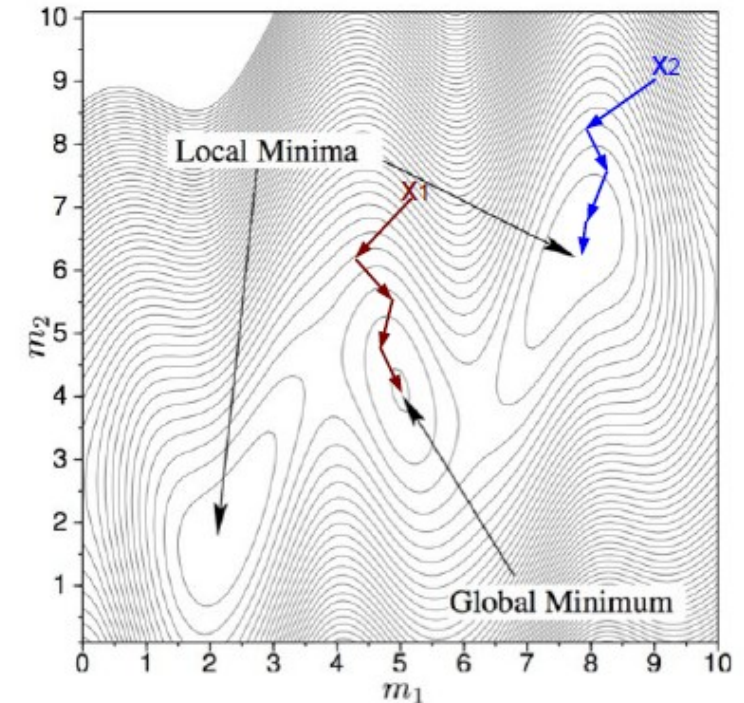
$$F = \frac{1}{2} \| \mathbf{d} - \mathbf{d}_{obs} \|^2 = \frac{1}{2} \| G\mathbf{m} - \mathbf{d}_{obs} \|^2 = \frac{1}{2} (\mathbf{d} - \mathbf{d}_{obs})^\dagger (\mathbf{d} - \mathbf{d}_{obs}) = \frac{1}{2} \Delta \mathbf{d}^\dagger \Delta \mathbf{d}$$

- Goal: find $\mathbf{m}$ that minimizes $F(\mathbf{m})$.
- Optimization is typically **iterative**, updating $\mathbf{m}$ step by step:

$$\mathbf{m}^{k+1} = \mathbf{m}^k + \Delta \mathbf{m}$$

where $\Delta \mathbf{m}$ is computed from the gradient of $F(\mathbf{m})$.

- Due to the presence of local minima, the starting point must be relatively close to the global minimum



a) local method



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Inverse Problem

$$F(m_0 + \Delta m) = F(m_0) + \sum_{j=1}^M \frac{\partial F(m_0)}{\partial m_j} \Delta m_j \\ + \frac{1}{2} \sum_{j=1}^M \sum_{k=1}^M \frac{\partial^2 F(m_0)}{\partial m_j \partial m_k} \Delta m_j \Delta m_k + \mathcal{O}(m^3)$$

- The gradient vanishes in the minimum position

$$\frac{\partial F(m)}{\partial m_l} = \frac{\partial F(m_0)}{\partial m_l} + \sum_{j=1}^M \frac{\partial^2 F(m_0)}{\partial m_j \partial m_l} \Delta m_j$$

- Rearranging the terms,

$$\Delta m = - \left(\frac{\partial^2 F(m_0)}{\partial m^2} \right)^{-1} \frac{\partial F(m_0)}{\partial m}$$



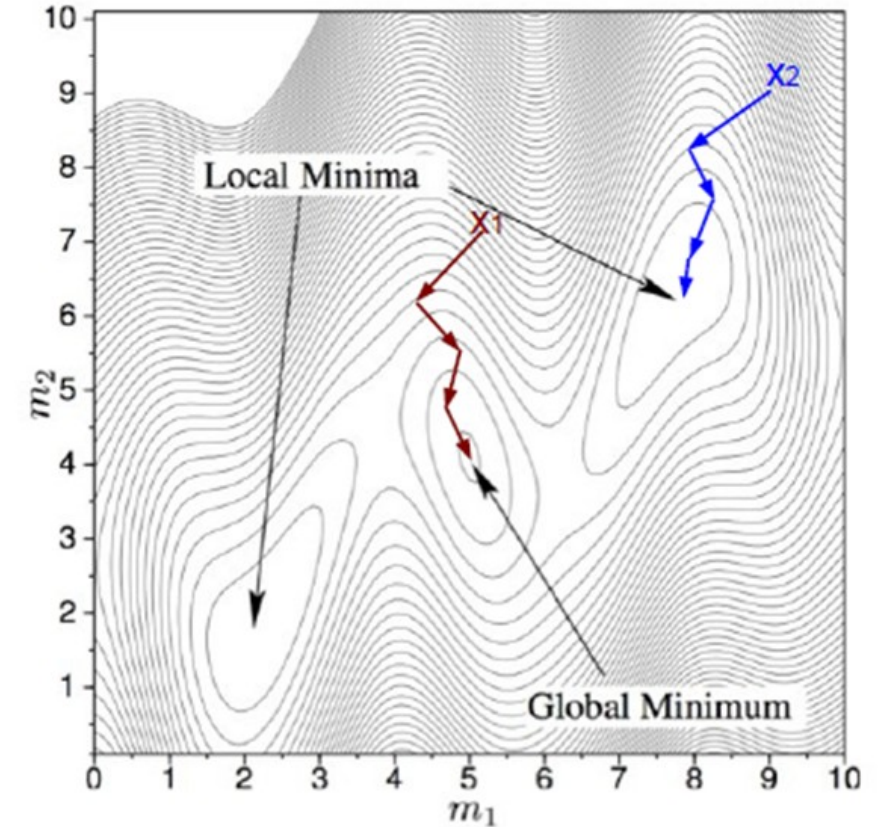
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Gradient and Hessian

$$\Delta m = - \left(\frac{\partial^2 F(m_0)}{\partial m^2} \right)^{-1} \frac{\partial F(m_0)}{\partial m}$$

- **1. Gradient (First-order derivative)**
 - Points in the **direction of steepest ascent** of the misfit function.
 - Model is updated in the **opposite direction** (steepest descent).
 - Guides the inversion towards lower misfit.
- **2. Hessian (Second-order derivative)**
 - Defines the **curvature** of the misfit function at m_0 .
 - For convergence to a minimum, the Hessian must be **positive definite**.
- **3. Practical considerations**
 - Full Hessian computation is **computationally expensive** (memory + time).
 - Common approximations:
 - **Gauss-Newton** and **quasi-Newton** methods (Pratt et al., 1998).
 - **Steepest-descent** method replaces the Hessian by a scalar for simplicity.



a) local method



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Iterative Optimization Methods

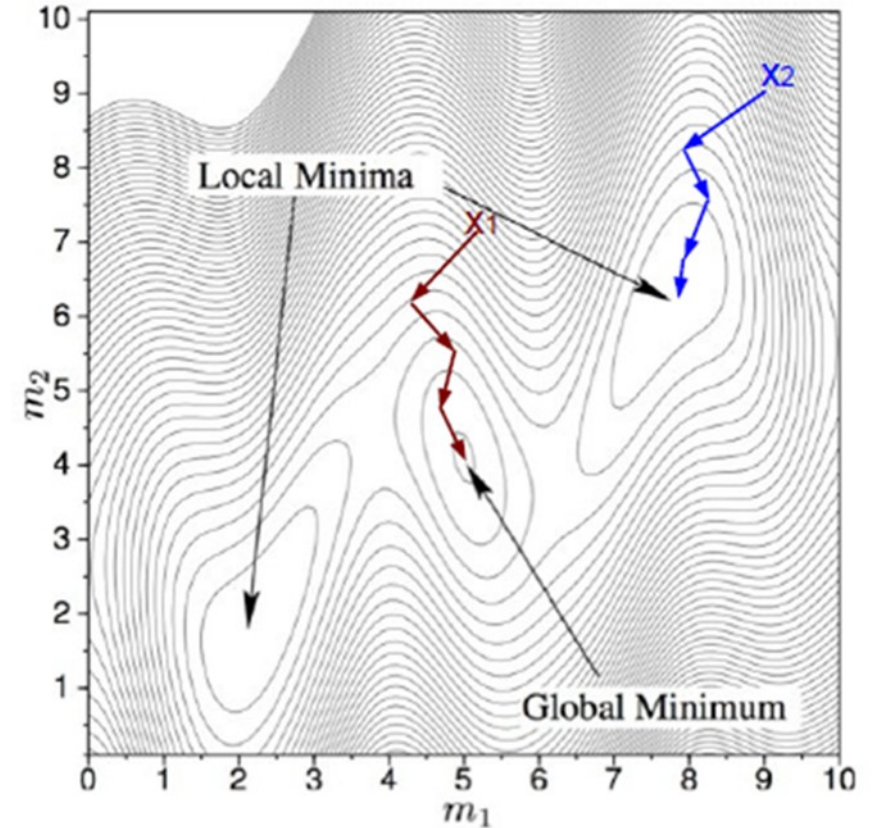
- **Steepest Descent:**

$$\Delta \mathbf{m} = \alpha \nabla F = -\alpha \Re \left[\left(\frac{\partial \mathbf{d}}{\partial \mathbf{m}} \right)^\dagger (\mathbf{d} - \mathbf{d}_{obs}) \right]$$

Moves opposite to gradient direction. Simple but slow.

- **Conjugate Gradient (CG):** Faster convergence by orthogonalizing directions.
- **Newton's Method:** Uses 2nd derivatives (Hessian) for curvature correction.
- **Quasi-Newton (L-BFGS):** Approximates Hessian efficiently for large-scale problems.

Key idea: Iterate until $F(\mathbf{m})$ (data misfit) no longer decreases significantly.

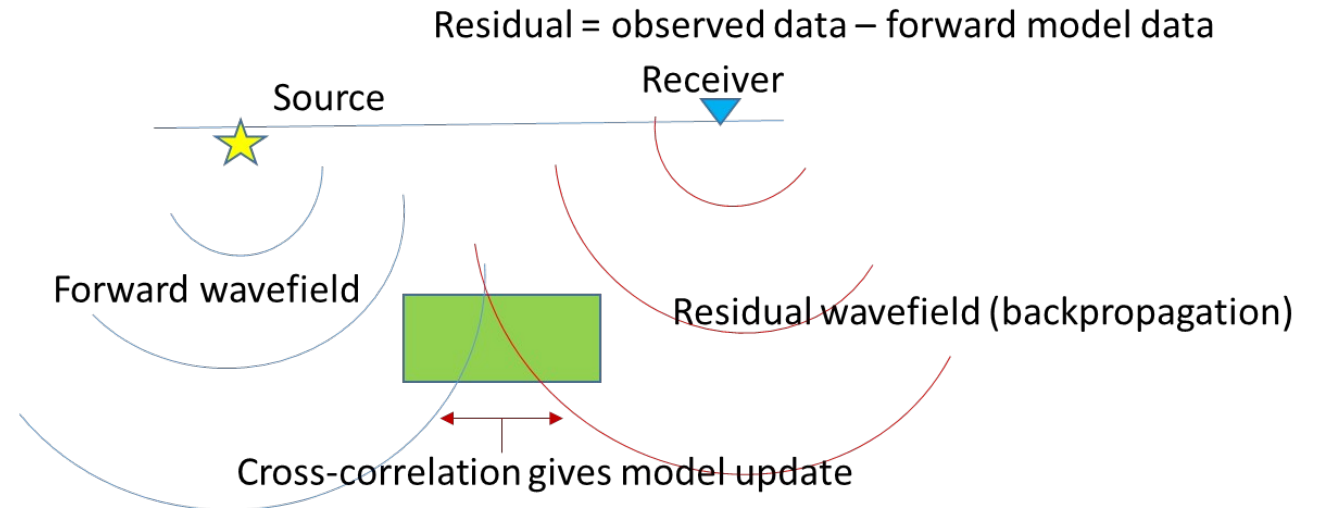


a) local method

Adjoint State Method

$$\Delta \mathbf{m} = \alpha \nabla F = -\alpha \Re \left[\left(\frac{\partial \mathbf{d}}{\partial \mathbf{m}} \right)^\dagger (\mathbf{d} - \mathbf{d}_{obs}) \right]$$

- Too expensive to calculate the term $\left(\frac{\partial \mathbf{d}}{\partial \mathbf{m}} \right)^\dagger$
- This can be avoided using adjoint state method
- **Steps:**
 - **Forward modelling:** compute synthetic data d_{mod} .
 - **Residuals:** $r = d_{mod} - d_{obs}$.
 - **Adjoint modelling:** back-propagate residuals using the adjoint wave equation (refer to Plessix, 2016).
 - **Gradient:** cross-correlate forward and adjoint wavefields over time.
- **Result:** Gradient of $F(m)$ with respect to model parameters (e.g., V_P, V_S)



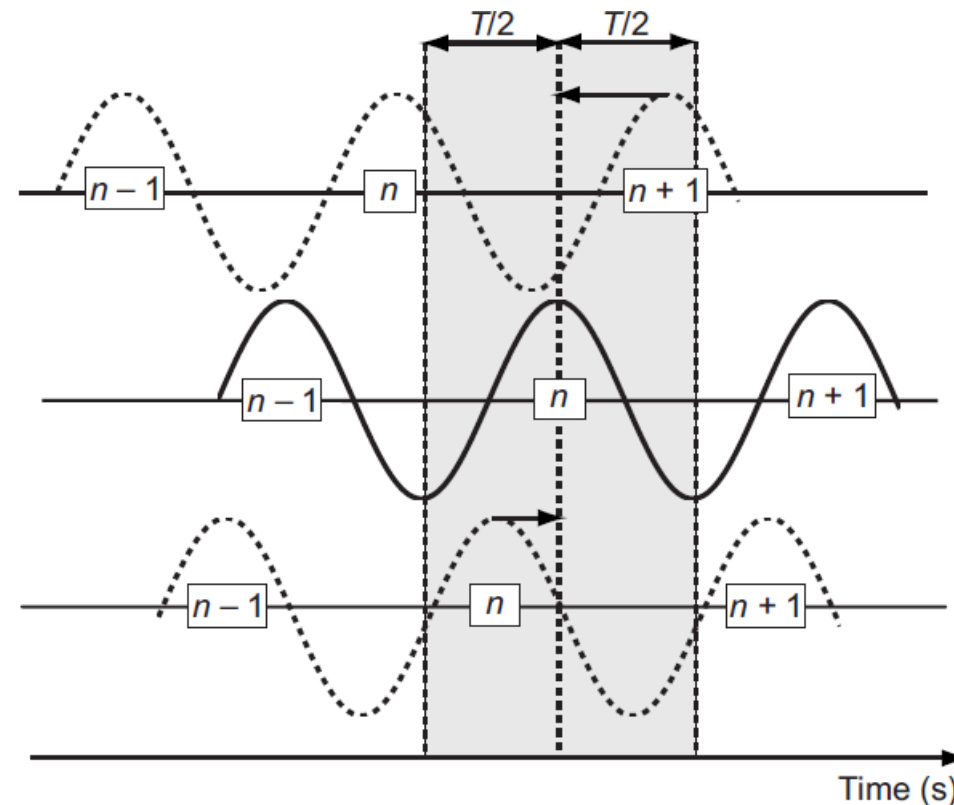


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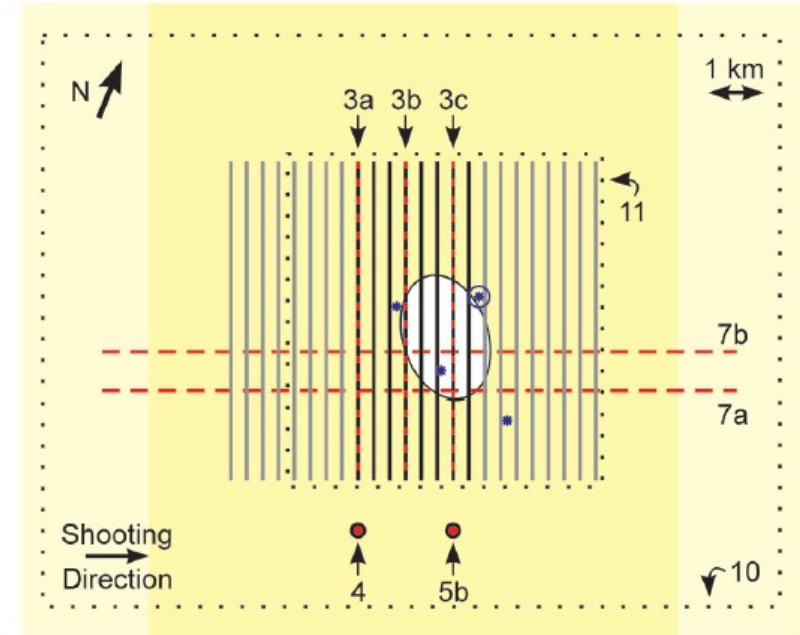
Challenges in Inversion

- **Cycleskipping**
- **Nonlinearity:** Perform the inversion small steps (iterate)
- **Non-uniqueness:** Requires starting model and more prior information will help
- **Computational cost:** 3D elastic FWI requires thousands of forward/adjoint simulations.
- **Strategies:**
 - Use hierarchical frequency bands.
 - Employ efficient parallelization and memory optimization.



Example: Tommeliten Field (Warner et al. 2013)

- Survey type: 3D, 4C (four-component), ocean-bottom-cable (OBC) survey.
- Geometry:
 - 3 swaths, each with 8 parallel cables (6 km each).
 - Receiver spacing: 25 m inline, 300 m crossline.
 - Sources: flip-flop shooting, two 3930 in³ airguns, 25 m × 75 m grid.



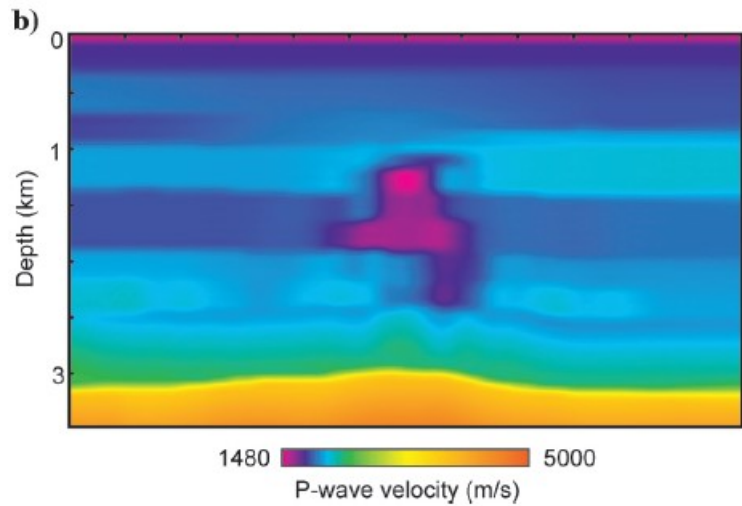
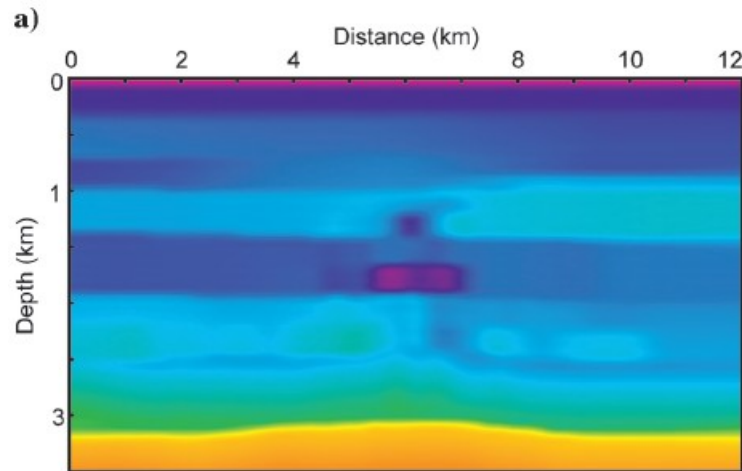
Experimental geometry of the 3D ocean-bottom-cable (OBC) survey across the Tommeliten Alpha Field. Eight black lines represent receiver cables in the central swath, which recorded sources within the **yellow source area**. Two additional swaths (grey cables and pale-yellow shot areas) extend coverage laterally. The **outer box** marks the limits of the FWI velocity model, and the **white zone** indicates the gas cloud causing seismic obscuration. **Blue dots** mark well locations



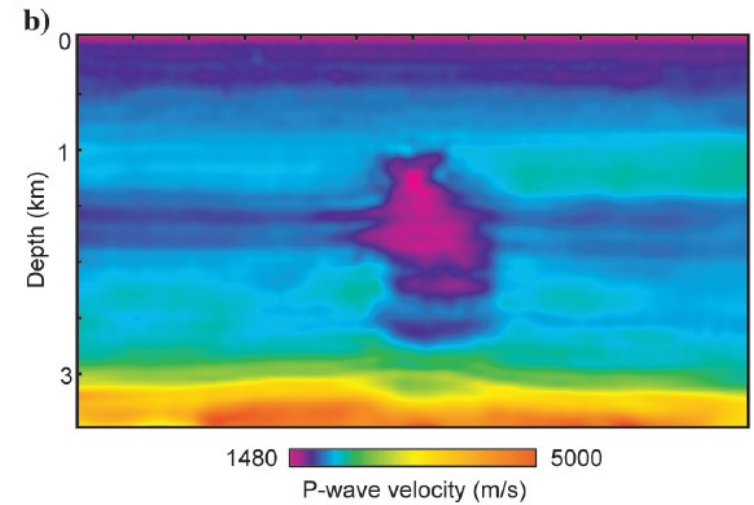
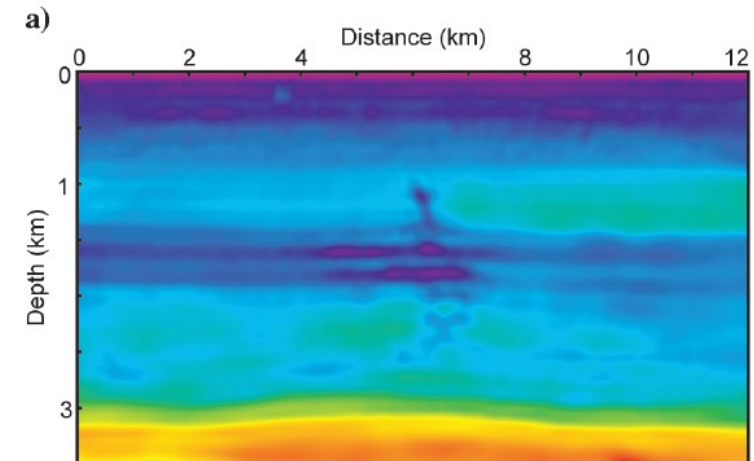
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FWI result



- Starting model



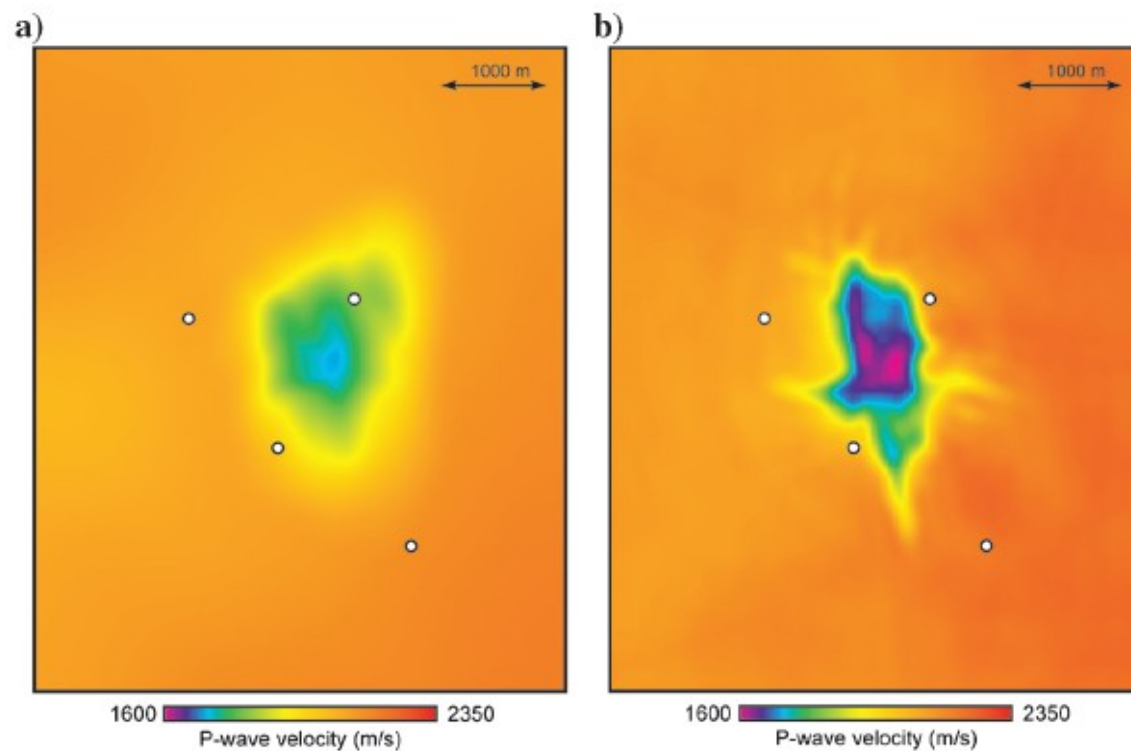
- FWI model



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FWI Result



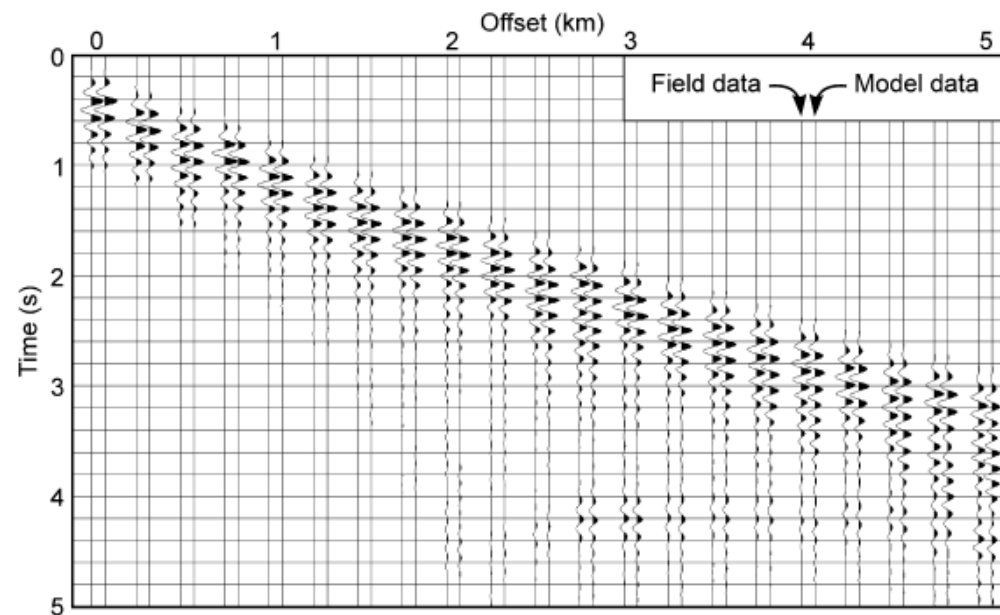
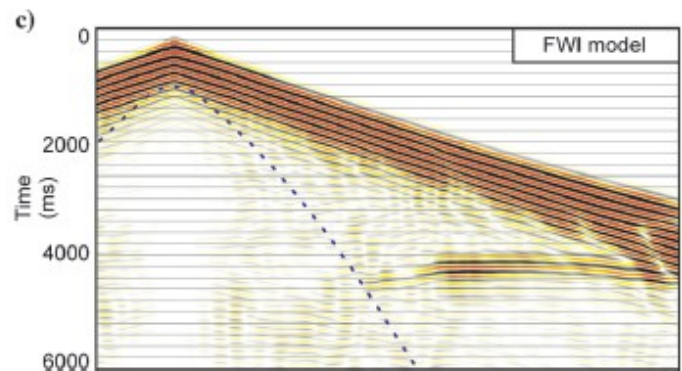
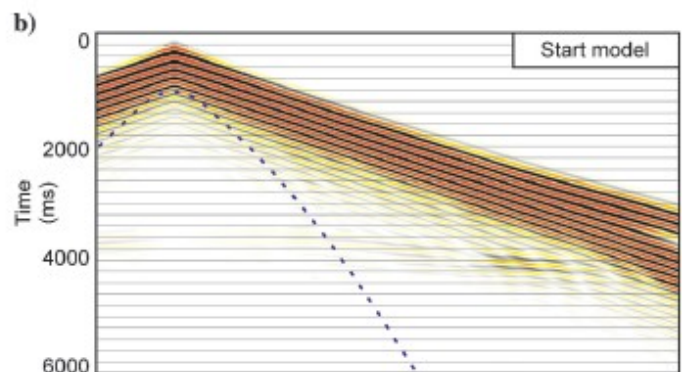
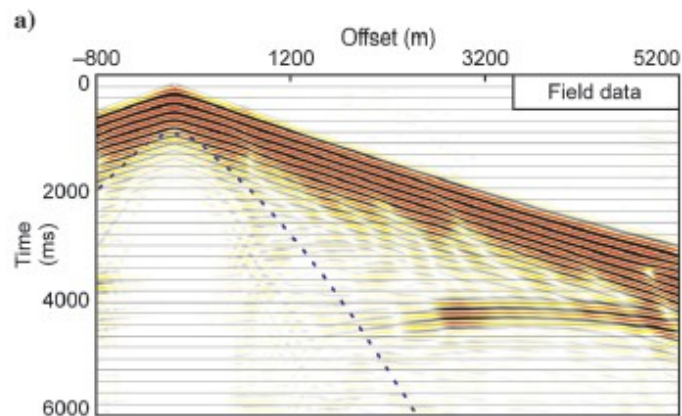
Horizontal slice from 1200 m depth starting and FWI models



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Data fit



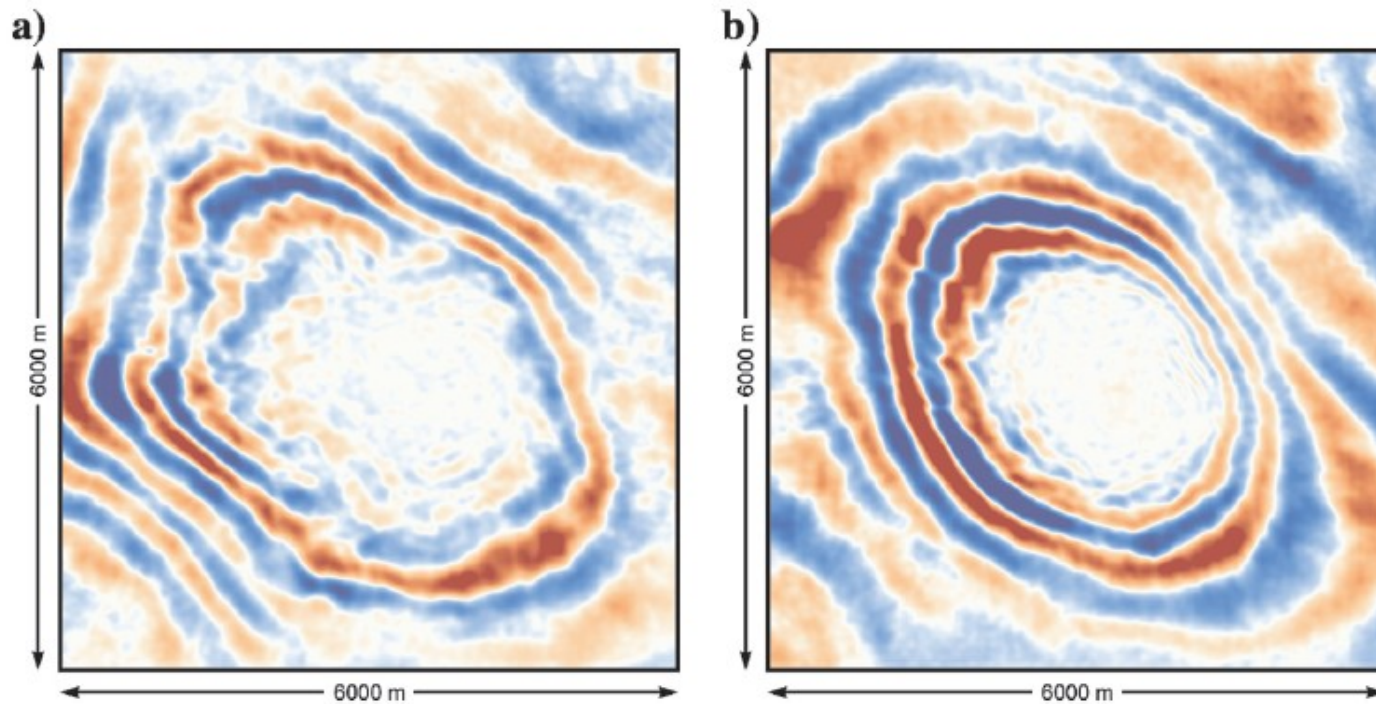
- Quality controlling the FWI result by matching the real and synthetic data
- FWI predicted data fits real data better



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Reverse time migrated images



- Left image is migration using starting model and right from FWI

75 år med teknologi for et bedre samfunn

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